

Modality-Agnostic Deep Clustering via Latent-Space Normalizing Flows for Medical Imaging

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Abstract

We propose a multi-modal deep clustering framework for medical imaging in which each modality (e.g., MRI, CT) retains its own VAE encoder and decoder, while a normalizing flow operating in the low-dimensional latent space bridges the modality-specific representations. The flow defines a shared modality-agnostic bottleneck $\mathbf{u}$ where a GMM prior is placed for clustering. One flow maps MRI latent codes to $\mathbf{u}$; a second flow maps CT latent codes to $\mathbf{u}$; both are invertible, enabling exact cross-modal generation in either direction. This design preserves the existing VAE architecture, confines the normalizing flow to a small latent dimension (e.g., 32), and provides an invertible bridge for cross-modal synthesis. We present the formulation, architecture, training protocol, and an evaluation plan for brain and cardiac imaging.

Keywords: Deep clustering, Cross-modal synthesis, Normalizing flows, Latent-space alignment

1 Introduction

Clinical phenotyping involves multiple imaging modalities. For instance, a patient may undergo MRI for tissue characterization, CT angiography for coronary anatomy, and electrophysiology (EP) mapping for arrhythmia substrates. Each modality captures a different aspect of the same pathophysiology, and a complete phenotype emerges only when these views are integrated.

Our starting point is a single-modality GMM-VAE that clusters images through a trainable Gaussian Mixture Model prior in the latent space (Section 2). Extending this to multiple modalities requires solving two problems: (1) aligning the latent representations so that the same disease phenotype maps to the same cluster regardless of input modality, and (2) enabling cross-modal image synthesis (e.g., MRI $\rightarrow$ CT) through the latent space.

Training separate models per modality and aligning their latent spaces post hoc is inadequate: independently learned latent geometries assign arbitrary cluster indices to the same phenotype, and decoders trained on one encoder’s output distribution cannot reliably decode latent codes from another. Replacing the entire encoder and decoder with normalizing flows addresses the decoder mismatch problem but sacrifices the flexible convolutional architectures that perform well on medical images.

We propose an intermediate approach. Each modality keeps its own VAE encoder and decoder, unchanged from the single-modality baseline. A normalizing flow operates only in the latent space (dimension D , e.g., 32), bridging the modality-specific latent codes through a shared representation $\mathbf{u}$ where a modality-agnostic GMM is placed. Because the flow is invertible, cross-modal translation is exact. Because the flow operates on D -dimensional vectors rather than images, it is small, fast, and does not constrain the encoder or decoder architecture.

2 Background

Let K be the number of clusters and D the latent dimensionality. The single-modality baseline assumes $\mathbf{z} \in \mathbb{R}^D$ drawn from a GMM prior:

$$p_{\theta}(\mathbf{z}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{z} \mid \boldsymbol{\mu}_k, \text{diag}(\boldsymbol{\sigma}_k^2)), \quad \sum_{k=1}^K \pi_k = 1, \quad (1)$$

with trainable parameters $\theta = \{\boldsymbol{\mu}_k, \log \boldsymbol{\sigma}_k^2, \text{logit}(\pi_k)\}_{k=1}^K$. A convolutional VAE is trained with a weighted loss:

$$\mathcal{L}_{\text{single}} = w_r \mathcal{L}_{\text{recon}} + w_{\text{KL}} \mathcal{L}_{\text{KL}} + w_g \mathcal{L}_{\text{GMM}}, \quad (2)$$

where $\mathcal{L}_{\text{GMM}} = -\log \sum_k \pi_k \mathcal{N}(\mathbf{z} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$. This formulation performs well for a single modality but does not support cross-modal alignment or generation.

3 Architecture

The proposed architecture (Figure 1) consists of three components: modality-specific VAE encoder/decoders, per-modality latent-space normalizing flows, and a shared modality-agnostic GMM.

Modality-specific VAEs. Each modality m has its own encoder E_m and decoder D_m with the same convolutional architecture as the single-modality baseline. E_{MRI} maps MRI images to $\mathbf{z}_{\text{MRI}} \in \mathbb{R}^D$; E_{CT} maps CT images to $\mathbf{z}_{\text{CT}} \in \mathbb{R}^D$. Each decoder reconstructs only its own modality.

Latent-space normalizing flows. Two normalizing flows connect the modality-specific latent spaces to a shared representation, operating entirely in $\mathbb{R}^D$. We define $h : \mathbb{R}^D \rightarrow \mathbb{R}^D$ as the mapping from MRI latent space to the shared representation $\mathbf{u}$, and $g : \mathbb{R}^D \rightarrow \mathbb{R}^D$ as the corresponding mapping from CT latent space. Both are composed of L invertible affine coupling layers [3] with ActNorm [4], using 2-hidden-layer MLPs (128 units, SiLU) for scale and translation networks.

The MRI side maps $\mathbf{u} = h(\mathbf{z}_{\text{MRI}})$, with inverse $\mathbf{z}_{\text{MRI}} = h^{-1}(\mathbf{u})$. The CT side maps $\mathbf{u} = g(\mathbf{z}_{\text{CT}})$, with inverse $\mathbf{z}_{\text{CT}} = g^{-1}(\mathbf{u})$. Cross-modal translation from MRI to CT is $g^{-1}(h(\mathbf{z}_{\text{MRI}}))$; from CT to MRI it is $h^{-1}(g(\mathbf{z}_{\text{CT}}))$.

Shared modality-agnostic GMM. The shared representation $\mathbf{u}$ is the modality-agnostic space where both flows converge:

$$\mathbf{u} = h(\mathbf{z}_{\text{MRI}}) = g(\mathbf{z}_{\text{CT}}).$$

A GMM prior (Equation (1)) is placed on $\mathbf{u}$:

$$p(\mathbf{u}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{u} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k). \quad (3)$$

Both modalities contribute to $\mathbf{u}$ through their respective flows, so the GMM components capture disease phenotypes rather than modality-specific appearance. Cluster assignment $k^* = \arg \max_k p(k \mid \mathbf{u})$ is modality-agnostic by construction.

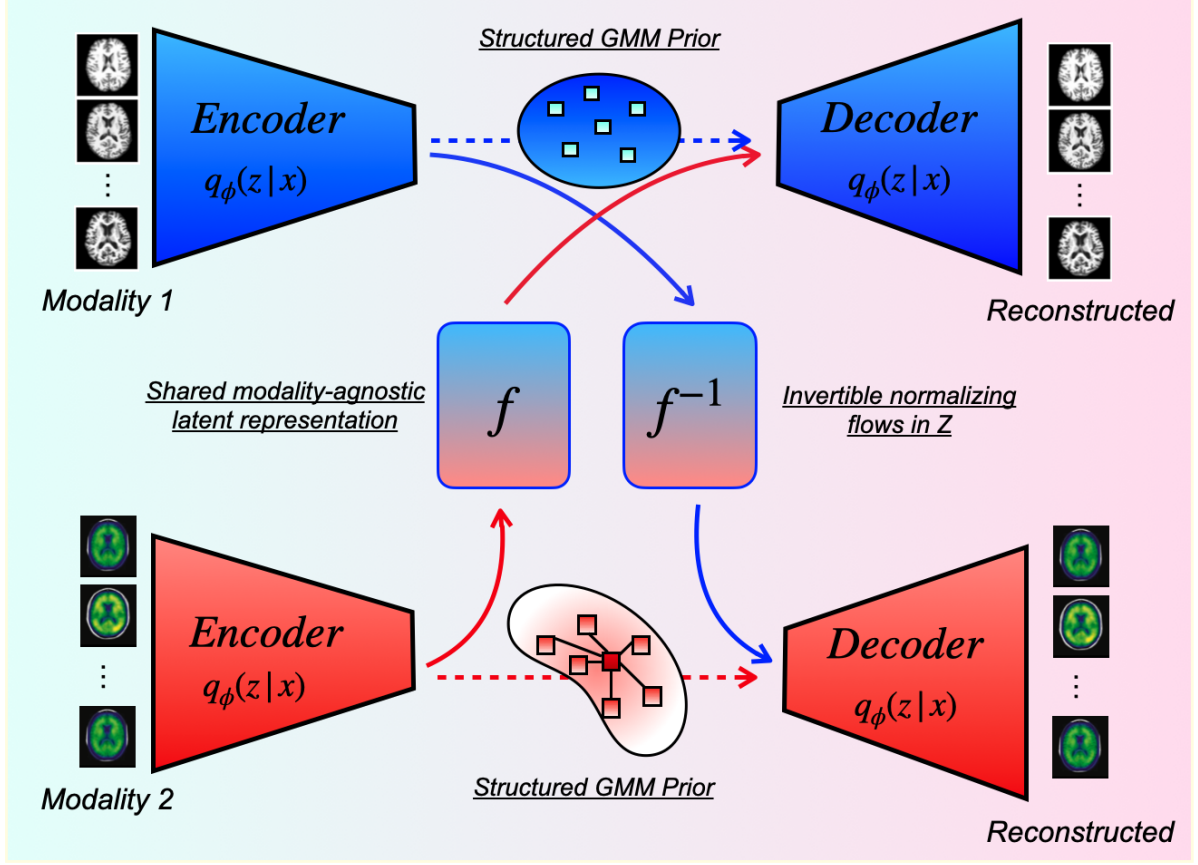


Figure 1: Proposed architecture. Each modality has its own VAE encoder $q_\phi(z|x)$ and decoder (blue: Modality 1, red: Modality 2), producing modality-specific structured GMM priors in the latent space. Invertible normalizing flows f and f^{-1} operate in the latent space, mapping both representations to and from a shared modality-agnostic latent representation (center). Self-reconstruction follows the dashed horizontal path within each modality. Cross-modal generation follows the curved arrows through the shared representation.

4 Cross-Modal Pathways

Self-reconstruction. An MRI image passes through its own encoder, flow, inverse flow, and decoder: $\mathbf{x}_{\text{MRI}} \rightarrow E_{\text{MRI}} \rightarrow \mathbf{z}_{\text{MRI}} \rightarrow h \rightarrow \mathbf{u} \rightarrow h^{-1} \rightarrow \hat{\mathbf{z}}_{\text{MRI}} \rightarrow D_{\text{MRI}} \rightarrow \hat{\mathbf{x}}_{\text{MRI}}$. Because $h^{-1} \circ h = \text{id}$, the flow introduces no distortion in the self-reconstruction path.

Cross-modal generation. MRI to CT: $\mathbf{x}_{\text{MRI}} \rightarrow E_{\text{MRI}} \rightarrow \mathbf{z}_{\text{MRI}} \rightarrow h \rightarrow \mathbf{u} \rightarrow g^{-1} \rightarrow \hat{\mathbf{z}}_{\text{CT}} \rightarrow D_{\text{CT}} \rightarrow \hat{\mathbf{x}}_{\text{CT}}$. The reverse direction uses g and h^{-1} . Each decoder receives latent codes within its own trained distribution, avoiding the distribution mismatch that arises when a decoder receives latent codes from a foreign encoder.

Cluster-conditioned generation. To generate a sample from cluster k in modality m : sample $\mathbf{u} \sim \mathcal{N}(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$, then apply the appropriate inverse flow (h^{-1} for MRI, g^{-1} for CT) followed by the corresponding decoder.

Clustering. For any input image $\mathbf{x}_m$, the encoder produces $\mathbf{z}_m$, the corresponding flow maps it to $\mathbf{u}$, and the cluster assignment is $k^* = \arg \max_k p(k|\mathbf{u})$. Because $\mathbf{u}$ is modality-agnostic, the assignment is consistent across modalities.

5 Training Objective

Per-modality VAE loss. Each modality is trained with its standard GMM-VAE objective, with the GMM loss computed at $\mathbf{u}$ rather than at $\mathbf{z}_m$:

$$\mathcal{L}_m^{\text{vae}} = w_r \|\mathbf{x}_m - D_m(\mathbf{z}_m)\|^2 + w_{\text{KL}} D_{\text{KL}}(q_m(\mathbf{z}|\mathbf{x}_m)\|p_0(\mathbf{z})) + w_g \mathcal{L}_{\text{GMM}}(\mathbf{u}_m), \quad (4)$$

where $\mathbf{u}_m$ is obtained by passing $\mathbf{z}_m$ through the appropriate flow, and $p_0(\mathbf{z})$ is a standard normal prior on the modality-specific latent space. The reconstruction and KL terms operate on $\mathbf{z}_m$; only the GMM term operates on $\mathbf{u}$.

Flow likelihood. The normalizing flow is trained by maximizing the likelihood of the mapping. For the MRI direction:

$$\mathcal{L}_{\text{flow}} = -\log p_{\text{CT}}(\mathbf{z}_{\text{CT}}) - \log |\det J_h(\mathbf{z}_{\text{MRI}})|, \quad (5)$$

where p_{CT} is the empirical distribution of CT encoder outputs.

Paired alignment. For paired data (same patient imaged in both modalities), we enforce agreement at the shared representation:

$$\mathcal{L}_{\text{pair}} = \frac{1}{|P|} \sum_{(i,j) \in P} \|h(\mathbf{z}_i^{\text{MRI}}) - g(\mathbf{z}_j^{\text{CT}})\|^2. \quad (6)$$

Cross-reconstruction. For paired data, we additionally train cross-modal reconstruction end-to-end:

$$\mathcal{L}_{\text{cross}} = \frac{1}{|P|} \sum_{(i,j) \in P} \left[\|\mathbf{x}_j^{\text{CT}} - D_{\text{CT}}(g^{-1}(\mathbf{u}_i))\|^2 + \|\mathbf{x}_i^{\text{MRI}} - D_{\text{MRI}}(h^{-1}(\mathbf{u}_j))\|^2 \right], \quad (7)$$

where $\mathbf{u}_i = h(\mathbf{z}_i^{\text{MRI}})$ and $\mathbf{u}_j = g(\mathbf{z}_j^{\text{CT}})$.

Total objective.

$$\boxed{\mathcal{L}_{\text{total}} = \sum_m w_m \mathcal{L}_m^{\text{vae}} + \lambda_f \mathcal{L}_{\text{flow}} + \lambda_p \mathcal{L}_{\text{pair}} + \lambda_{\text{cross}} \mathcal{L}_{\text{cross}} + \lambda_b \mathcal{L}_{\text{balance}}} \quad (8)$$

Training schedule. Phase 1 (epochs 1–10): train each VAE independently with its own GMM loss at $\mathbf{u}$; flow layers train alongside to map each $\mathbf{z}_m$ to $\mathbf{u}$; $\lambda_p = \lambda_{\text{cross}} = 0$. Phase 2 (epochs 11–30): activate paired alignment and cross-reconstruction, ramping λ_p and λ_{cross} linearly over 5 epochs. Phase 3 (epochs 31+): full objective with cosine learning rate annealing.

6 Atlas Building

The **native atlas** is computed per (cluster k , modality m) pair by collecting assigned samples at $\mathbf{u}$ and taking the robust pixel-space median, following the existing single-modality pipeline.

The **cross-modal atlas** decodes each GMM component mean through each modality’s inverse flow and decoder:

$$\hat{\mathbf{x}}_{\text{MRI}}^{(k)} = D_{\text{MRI}}(h^{-1}(\boldsymbol{\mu}_k)), \quad \hat{\mathbf{x}}_{\text{CT}}^{(k)} = D_{\text{CT}}(g^{-1}(\boldsymbol{\mu}_k)). \quad (9)$$

This produces a canonical representation of phenotype k in every modality.

7 Application: Cardiac Phenotyping

Table 1: Cardiac proof-of-concept setup.

Modality	Source	Preprocessing	D_m	D
Cardiac MRI (SA)	UK Biobank / ACDC	Resize 128×128, z-score	16 384	32
Cardiac CT	Public CT archives	Window [−150, 350] HU, resize	16 384	32
EP voltage maps	Clinical EP lab	Interpolate to 64×64 grid	4 096	32

Expected clusters correspond to normal myocardium, ischemic cardiomyopathy subtypes, hypertrophic cardiomyopathy, dilated cardiomyopathy, and arrhythmogenic substrates. MRI/CT pairs are available for approximately 15% of patients; MRI/EP pairs for approximately 5%.

Evaluation. Clustering quality: per-modality Hungarian accuracy, NMI, ARI, and cross-modal cluster agreement (CCA). Generation quality: FID and SSIM on cross-modal synthesis, and clinical expert review of cross-modal atlases. Key ablations: flow depth ($L \in \{4, 8, 12\}$), latent dimensionality ($D \in \{8, 16, 32\}$), paired fraction (100% to 0%), and cross-reconstruction on versus off.

8 Timeline and Expected Outcomes

Table 2: Projected timeline and deliverables.

Phase	Period	Deliverable
1	Weeks 1–3	Single-modality VAE baseline on cardiac MRI and CT separately. Verify clustering metrics (Hungarian accuracy, NMI, ARI) per modality (FINISHED).
2	Weeks 3–5	Implement latent-space normalizing flows h and g . Train with paired alignment loss on co-registered MRI/CT subset. Report cross-modal cluster agreement (CCA).
3	Weeks 5–7	Add cross-reconstruction loss. Generate cross-modal atlases. Evaluate FID/SSIM on synthesized images.
4	Weeks 7–8	Ablation studies (flow depth, latent dim, paired fraction). Clinical expert review of cross-modal atlases.

The primary expected outcome is a unified clustering that assigns consistent cluster labels across MRI, CT, and EP modalities without post-hoc alignment. Secondary outcomes include cross-modal image synthesis (e.g., generating a synthetic CT from an MRI input) and cross-modal atlases that visualize each disease phenotype in every imaging modality.

If the latent-space normalizing flow does not sufficiently resolve distribution mismatch between modalities, a fallback option is to supplement the flow with domain-adversarial training on $\mathbf{u}$, using a small discriminator to further remove residual modality information. An additional modality (e.g., EP voltage maps) can be incorporated by adding a third flow without modifying the existing components.

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